CLASS 9 — MATHS · FORMULA SHEET

Chapter 13: **Two Variables, One Line** | Quick revision, one page

Linear equation in two variables

- Standard form: $ax + by = c$ with a and b not both zero.
- A solution is a pair (x, y) that satisfies the equation; there are infinitely many solutions.
- Each solution is a point on the line; every point of the line is a solution.

Graph and intercepts

- The graph is a straight line; two points are enough to draw it (a third checks it).
- x-intercept: put $y = 0$, giving $\left(\frac{c}{a}, 0\right)$. y-intercept: put $x = 0$, giving $\left(0, \frac{c}{b}\right)$.
- Slope of $ax + by = c$ (with $b \neq 0$): $-\frac{a}{b}$; rewrite as $y = -\frac{a}{b}x + \frac{c}{b}$.

Special lines

- $x = k$: vertical line parallel to the y-axis. $y = k$: horizontal line parallel to the x-axis.
- $y = mx$: a line through the origin with slope m; $ax + by = 0$ also passes through the origin.
- Slope-intercept form: $y = mx + c$ has slope m and y-intercept c.

Using equations

- Translate the situation: total = (rate $\times$ number) + (rate $\times$ number).
- To test a point, substitute: it lies on the line only if both sides are equal.

MASTERMIND